

Selecting Optimal Unrestricted Hartree–Fock Trial Wave Functions for Phaseless Auxiliary-Field Quantum Monte Carlo: Accuracy and Limitations in Modeling Three Iron–Sulfur Clusters

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Published August 25, 2026

Abstract

Phaseless auxiliary-field quantum Monte Carlo (ph-AFQMC) has emerged as a promising electronic structure method for correlated electronic systems. However, the quality of its predictions depends critically on the choice of trial wave function $|\Psi_T\rangle$, and it is not obvious how to make an optimal choice, especially for strongly correlated states of large transition-metal complexes. Mean-field wave functions are compelling trial wave function candidates, as they map directly to chemical concepts and can be obtained with $O(N^4)$ cost. Yet in the strongly correlated regime, one faces a symmetry dilemma and the existence of multiple nearly degenerate broken-symmetry solutions. In this work, we investigate active-space models of $[2\text{Fe}-2\text{S}]^{2+}$, mixed-valent $[4\text{Fe}-4\text{S}]^{2+}$, and $[4\text{Fe}-4\text{S}]^{4+}$ and explore the sensitivity of ph-AFQMC to the choice of unrestricted Hartree–Fock (UHF) trial wave functions. We find that ground-state energies are highly sensitive to the spin contamination and local spin alignments of $|\Psi_T\rangle$, exposing fundamental limitations in classical mean-field trial states.

1 Introduction

Iron–sulfur ($[Fe-S]$) clusters are essential catalytic cofactors in metalloenzymes such as nitrogenase, ferredoxins, and respiratory complexes. Their electronic structure is defined by open-shell iron centers with strong magnetic exchange coupling and severe static electron correlation.

While exact diagonalizations like Full Configuration Interaction (FCI) scale exponentially, Phaseless Auxiliary-Field Quantum Monte Carlo (ph-AFQMC) achieves polynomial scaling by projecting the trial state $|\Psi_T\rangle$ in imaginary time:

$$|\Phi_0\rangle \propto \lim_{\tau \rightarrow \infty} e^{-\tau \hat{H}} |\Psi_T\rangle \quad (1)$$

The phaseless constraint controls the sign/phase problem by constraining random walks in Slater determinant space to regions where the overlap with $|\Psi_T\rangle$ satisfies $\text{Re}\{\langle\Psi_T|\phi\rangle\} > 0$. Consequently, the accuracy of ph-AFQMC is fundamentally bounded by the nodal surface of $|\Psi_T\rangle$.

2 The Mean-Field Symmetry Dilemma

Unrestricted Hartree–Fock (UHF) minimizes the mean-field energy by breaking spin and spatial symmetries:

$$\mathbf{F}^\alpha \mathbf{C}^\alpha = \mathbf{S} \mathbf{C}^\alpha \epsilon^\alpha, \quad \mathbf{F}^\beta \mathbf{C}^\beta = \mathbf{S} \mathbf{C}^\beta \epsilon^\beta \quad (2)$$

In $[Fe-S]$ clusters, UHF yields a proliferation of local minima with different local spin states $\langle S_z(Fe) \rangle$. The resulting trial states exhibit severe spin contamination:

$$\langle \hat{S}^2 \rangle_{\text{UHF}} \gg S(S+1) \quad (3)$$

This creates an acute trial state selection problem: does minimizing the mean-field energy E_{UHF} correlate with minimizing the ph-AFQMC phaseless bias?

3 Results on Iron–Sulfur Active Spaces

We compute the ground-state energies and vertical spin-state splittings for active-space models of $[2\text{Fe}-2\text{S}]^{2+}$, $[4\text{Fe}-4\text{S}]^{2+}$, and $[4\text{Fe}-4\text{S}]^{4+}$.

Finding 1 (Trial State Sensitivity). We observe that different UHF solutions within a 20 m E_h window can produce ph-AFQMC energy variations exceeding 15 kcal/mol. The lowest-energy UHF determinant is frequently *not* the optimal trial state.

Finding 2 (Broken-Symmetry vs. Entropic Trapping). Mean-field self-consistent-field (SCF) methods become trapped in broken-symmetry minima because the integer Aufbau principle forces sharp single-determinant projections that cannot continuously rotate through spatial degeneracies.

4 Conclusions

Our benchmarks demonstrate that while ph-AFQMC with single UHF determinants can achieve chemical accuracy for certain spin alignments, the manual search for the optimal mean-field trial wave function is intractable for complex multimetallic clusters. This underscores the need for alternative single-reference formalisms that regularize mean-field singularities without catastrophic symmetry breaking.

References

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